NOTE: The document identifier and heading have been changed on this page to reflect that this is a performance specification. There are no other changes to this document. The document identifier on subsequent pages has not been changed, but will be changed the next time this document is revised.

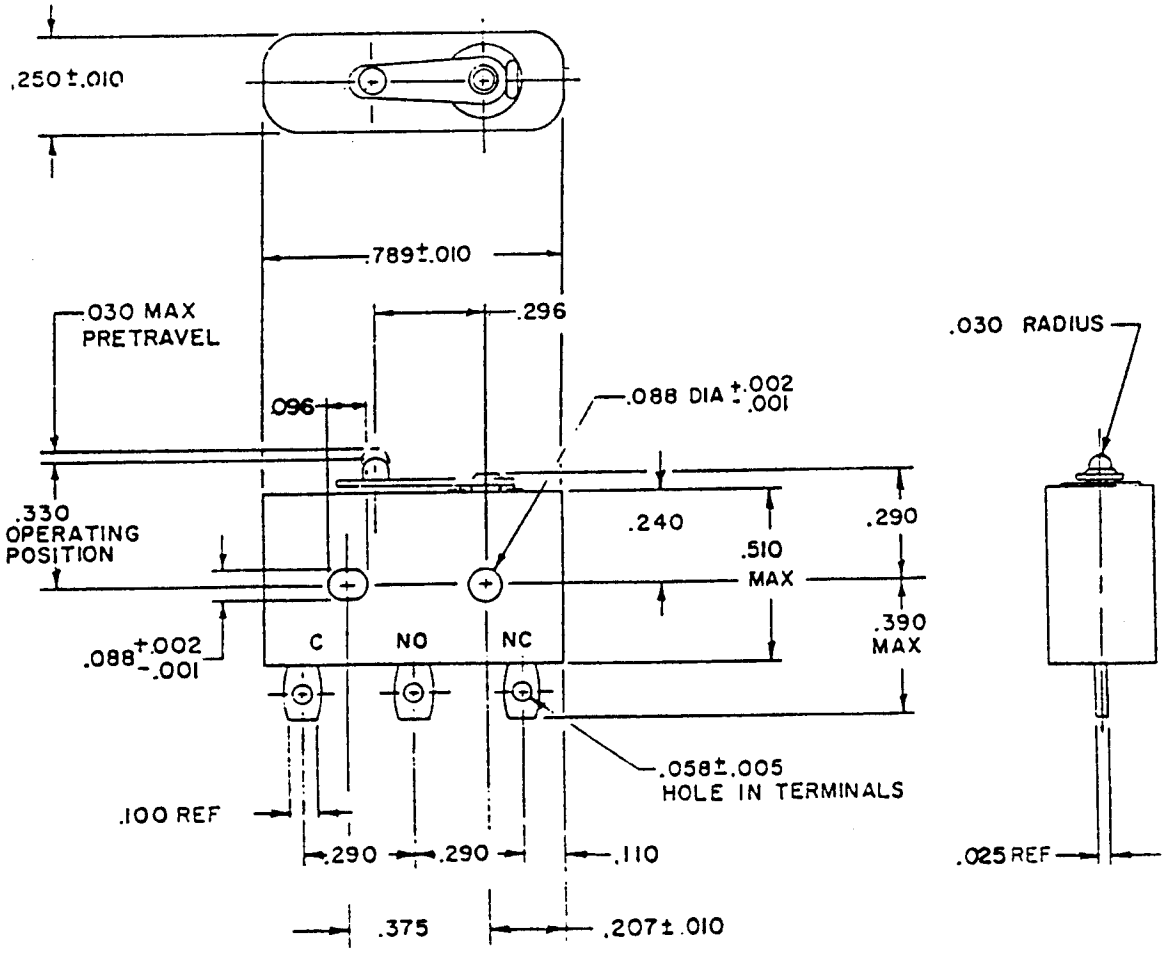
MIL-PRF-8805/8E
24 August 1978
SUPERSEDING
MIL-S-8805/8D
15 May 1970

PERFORMANCE SPECIFICATION SHEET

SWITCHES, SENSITIVE, SPDT, LEAF
ACTUATOR, LOW LEVEL TO 4 AMPS, HERMETIC SEAL

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the switches described herein shall consist of this document and the latest issue of Specification MIL-S-8805.



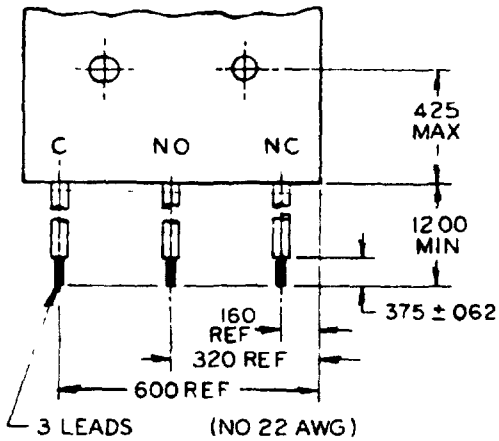
MS 27216-1, -5 & -6

FIGURE 1. Dimensions and configurations.

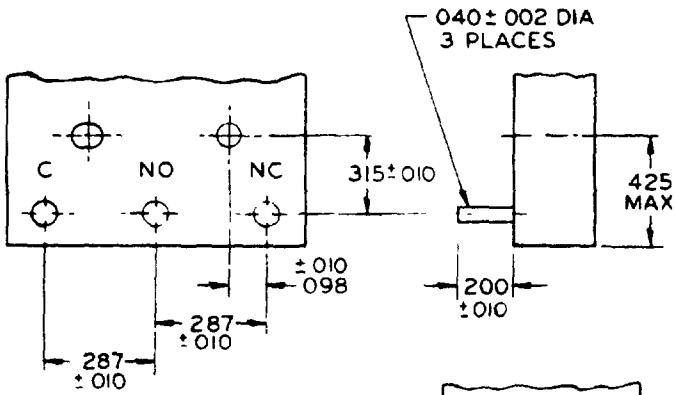
(E) denotes changes

FSC 5930

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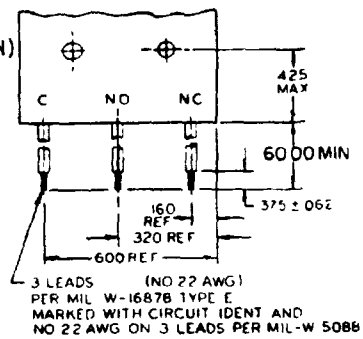


MS 27216-2



Ⓔ MS27216-3 (INACTIVE FOR NEW DESIGN)

INCHES	MM	INCHES	MM	INCHES	MM
001	03	098	2 49	315	8.00
002	05	100	2 54	320	8 13
005	13	110	2 79	330	8 38
010	25	160	4 06	375	9 52
025	64	200	5.08	390	9 91
030	76	207	5 26	425	10 80
040	1 02	240	6 10	510	12 95
058	1 47	250	6 35	600	15 24
062	1 57	287	7 29	789	20 04
088	2 24	290	7 37	12 00	304 8
096	2 44	296	7 52	60 00	1524 0



MS 27216-4

NOTES
1. Dimensions are in inches
2. The terminal designation shall be permanently and legibly marked on switch enclosure
3. Exact shape of switch optional, provided dimensions specified are not exceeded
4. Unless otherwise specified, tolerance is ± 015 (38 mm)
5. Mounting holes will accept pins or screws of 087 (2 21 mm) maximum diameter on 375 ± 002 (9 53 ± 05 mm) centers
6. Metric equivalents are given for general information only and are based upon 1 00 inch = 25 4 mm

FIGURE 1 Dimensions and configurations - Continued

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REQUIREMENTS

Dimensions and configuration: See figure 1

Enclosure design 5 (Hermetic)

Temperature characteristic 2 (-65° to +125°C).

(E) Shock type
MS27216-1, -2, -3, -4 M (100 g)
MS27216-5, -6 H (high impact)

Vibration grade 2 (10 to 2,000 Hz)

Weight
(E) MS27216-1, -5, -6 014 pound maximum.
MS27216-2 .026 pound maximum
MS27216-3 .020 pound maximum
MS27216-4 090 pound maximum

Operating characteristics
Actuating force 7 ounces maximum
Release force 1 ounce minimum
Movement differential 006 inch maximum
Pretravel 030 inch maximum
Overtravel 003 inch minimum

Terminal strength See table I

Dielectric withstanding voltage
(E) At atmospheric pressure
MS27216-1, -2, -3, -4, -5 1000 V rms
MS27216-6 800 V rms
At reduced barometric pressure 400 V rms
Following electrical endurance In qualification inspection table, the dielectric withstanding voltage points of application between all unconnected terminals of the same pole shall be tested at 200 V rms minimum
Contact resistance
MS27216-1 through -5 Not applicable.
(E) MS27216-6 50 milliohms maximum initial, 100 milliohms maximum after mechanical endurance.

Mechanical endurance 25,000 cycles

Electrical endurance 25,000 cycles

Electrical ratings See table II

(E) Low level circuit Applicable to MS27216-6

TABLE I. Terminal strength requirements

Part number	Applied force
(E) MS27216-1, -5, -6 - -	9 lbs
MS27216-2, -3, -4 - -	10 lbs

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TABLE II Electrical ratings

Load	Sea level		70,000 feet
	28 Vdc	115 Vac, 400 Hz	28 Vdc
	amperes	amperes	amperes
MS27216-1, -2, -3, -4			
Resistive Inductive Lamp	4 2 1	4 2 .5	4 2 1
(E) MS27216-5			
Resistive Inductive	3 1	1 1	3 1
(E) MS27216-6			
Resistive Inductive	0 5 0 25	--- ---	0.5 0 25

Qualification

Group submission

Sample The number of sample units for each part number covered by this specification sheet shall be as specified in table III

Test routine All sample units shall be subjected to visual and mechanical examination The sample units shall then be divided as specified in table III and subjected to the inspections indicated

(E) TABLE III Qualification inspection

Switches for testing	Number of sample units	Examination or test	Extent of approval
MS27216-1	30	Complete qualification table, as applicable, of MIL-S-8805	MS27216-1 MS27216-2 MS27216-4 MS27216-5 MS27216-6
MS27216-2	2	Insulation resistance Operating characteristics Terminal strength Vibration Shock Operating characteristics Visual and mechanical examination Seal	
MS27216-4	2	Visual and mechanical examination	
MS27216-5	24	Complete qualification table, as applicable, of MIL-S-8805 except for terminal strength, solderability, explosion	
MS27216-6	6	Visual and mechanical examination Seal Dielectric withstanding voltage Insulation resistance Operating characteristics Overload Electrical endurance Low level circuit Dielectric withstanding voltage Operating characteristics	

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Group C inspection.

Explosion test	Not applicable

Custodians
Army - EL
Navy - EC
Air Force - 85

Review activities
Army - AV, MI
Navy - SH, AS, OS
Air Force - 17, 99
DLA - ES

User activities
Army - AR
Navy - MC
Air Force - 19

Preparing activity:
Navy - EC

Agent
DLA - ES

(Project 5930-1053-10)

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NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

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A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

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